

HDR X GRUNDFOS · WASTE HEAT REUSAGE

Thermal Commons

Heat for Lansing: turning Lake Hawkeye's waste heat into a reason to say yes

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The heat is free. The town still might say no.

778 GWh

of heat a year leaves the 150 MW
first phase through dry coolers

No gas

Many homes burn propane and
oil; NYSEG has barred new gas
connections since Feb 2015 (2026
status unverified)

A ban

The Town Board told its attorney
to draft a data-center ban

Lansing, Michigan, 6 April 2026: Deep Green withdrew a heat-reuse data center after local opposition. Free heat was not enough.

OUR ANSWER

The conditions under which Lansing could say yes

Binding contracts

Community Benefit Agreement plus a Heat Supply Agreement with step-in rights

Community ownership

A thermal co-op owns the network; the data center supplies heat, not control

Public proof

An open dashboard reports heat delivered, prices and CO2 every month

The right tool at every density

Ring	Who gets heat	Cost of heat	Verdict
1 · On-site campus	10 ha greenhouse, aquaculture, pool	\$41 / MWh	Build
2 · Corridor	500 homes on an ambient loop	\$286 / MWh	Build with CBA funds
3 · Town center	Conditional hot main	\$734 / MWh	Not yet: fails our test
Scattered homes	Heat-pump rebates, no pipe	Rebate	Fund from CBA

Pipe goes only where heat density pays for it. Everyone else gets a heat pump, paid from the same community fund.

How it works: every hour of the year

Capture

Sidestream heat exchanger on the sealed glycol loop. Dry coolers stay 100% primary.

Upgrade

Central heat pump lifts to a 55-65°C hot loop. COP from real temperatures, bounded 2-6.

Store

Thermal storage shaves winter peaks; backup boilers stay at anchors.

Price

Capex, opex, electricity, 30-year capital recovery, tariff vs propane and oil.

8,760-hour simulation on Ithaca typical-year weather, in an open Python model with tests. The web app recalculates the economics live as you move the sliders.

$$\text{COP} = 0.5 \cdot T_{\text{sink}} / (T_{\text{sink}} - T_{\text{source}} + 6 \text{ K})$$

What it delivers, base case

50.6 GWh

of heat delivered a year to the farm campus and 500 homes

\$106 / MWh

blended cost of heat vs propane \$136 and oil \$156

11,408 t

CO2 avoided a year, about 2,659 cars off the road

\$735 / yr

saved per household vs propane (\$1,266 vs oil)

THE HONEST PART

Pipe to scattered homes does not pay on its own. The community fund closes the gap.

~1.7%

of data-center capex funds the corridor gap and heat-pump rebates, through the Community Benefit Agreement

The on-site campus pays for itself.

A town-center main fails our own test, so we do not build it.

If the data center leaves, step-in rights and storage protect the co-op.

What Lansing gets

500

homes off propane and oil

126

local jobs

5,500 t

of food a year from a 10 ha
greenhouse

1,500 t

of fish a year, plus a year-
round pool

Scored against HDR's seven domains: health, community, air, carbon, water, biodiversity and nutrients. We do not claim the plan saves lake water one-for-one.

Don't ban it. Set the terms.

Sign a Community Benefit Agreement and a Heat Supply Agreement, start with the on-site farm campus, and let a community co-op own the heat.

Live demo next: move the sliders and watch the economics recalculate.